

Ultimate Nutrition

ATP CHARGE

Adenosine Tri-phosphate (or simply ATP) is the energy-rich chemical compound that provides virtually all the energy needed by your body. The energy released from the breakdown of ATP is used to power all body functions. We need ATP to make our hearts beat, to give our muscles power when we demand it, and to maintain our everyday lives. Without adequate ATP stores, we could not walk, run, breathe, or even have blood flow through our bodies. So, ATP is considered the "energy currency" of the cell. It is, in fact, the molecule that gives us life.

Our bodies must be continuously supplied with energy to perform its many complex functions. As the body's energy demands increase with exercise, there must be a way to provide this additional energy so that we can continue functioning at our optimum levels.

Adenosine Triphosphate (ATP) is the basic fuel used by all cells in the body. The force of all muscle contraction is a result of the chemical conversion of ATP. ATP stores virtually all of the energy needed by the body. During the normal process of metabolism, ATP is hydrolyzed—broken down with water and oxygen. The process of break down causes a burst of energy, giving ATP its "energy currency" status because it releases vital energy within cells countless times throughout the day. This energy keeps the heart beating and oxygen flowing and empowers muscle cells to meet daily demands.

The body's store of ATP is very limited and must be constantly manufactured to keep up with energy demand. ATP is the primary source for muscle contractions. Whenever we use our muscles, ATP is broken down into two simpler chemicals (adenosine *di*-phosphate - ADP and inorganic phosphate). This process of ATP turning into ADP releases the energy which gives our muscles power. The problem is that our muscles only contain enough ATP to last about 10-15 seconds at maximum exertion. Muscle contractions place a demand on ATP resulting in insufficient amounts necessary for protein synthesis.

Creatine Phosphate or Phosphocreatine (the major phosphorylated compound in muscle tissue) is the back up source for ATP. Phosphocreatine declines rapidly in the muscle tissue during physical activity, which can also diminish the amounts of ATP, causing extreme fatigue. Increased ATP along with creatine may enable more muscle contractions resulting in longer and more vigorous workouts. This can enhance sports activity and performance.

Ultimate Nutrition understands that getting ATP into the body can be a difficult task with oral supplementation because ATP is mostly, if not completely destroyed in the digestive system. That's why we offer sublingual (under-the-tongue) ATP Charge, which will bypass the digestive system.

SELECTED REFERENCES:

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FREQUENTLY ASKED QUESTIONS:

1. What is ATP?

ATP, or Adenosine Triphosphate, is a molecule found in mitochondria, and is the energy source of all cells. It is converted to ADP (Adenosine Diphosphate), which is the energy necessary for transmitting nerve cells, muscle contractions, and cell division. As part of the Krebs cycle, it is involved in the metabolic process of amino acid and carbohydrate metabolism.

2. Why supplement with ATP?

Since adenosine triphosphate (ATP) is so important, you might think that our cells would stockpile the stuff big-time, but that is not the way it works. Our cells, including our muscle cells, refuse to store large quantities of the precious high-energy phosphate, being much more interested in storing carbohydrates and fat. Because of this, and since muscular exercise depends on a steady, often expansive, supply of adenosine triphosphate (ATP) to provide the energy needed for muscular contractions, muscles need dependable 'metabolic pathways', which can provide adenosine triphosphate (ATP) at a rapid, reliable rate.

3. Why is the sublingual form of ATP so popular?

Getting ATP into the body is a difficult task with oral supplementation because ATP is mostly, if not completely destroyed in the digestive system. However, a sublingual (under-the-tongue) form of ATP will bypass the digestive system. The sublingual delivery system allows for faster uptake into the bloodstream, resulting in better absorption and increased effectiveness.